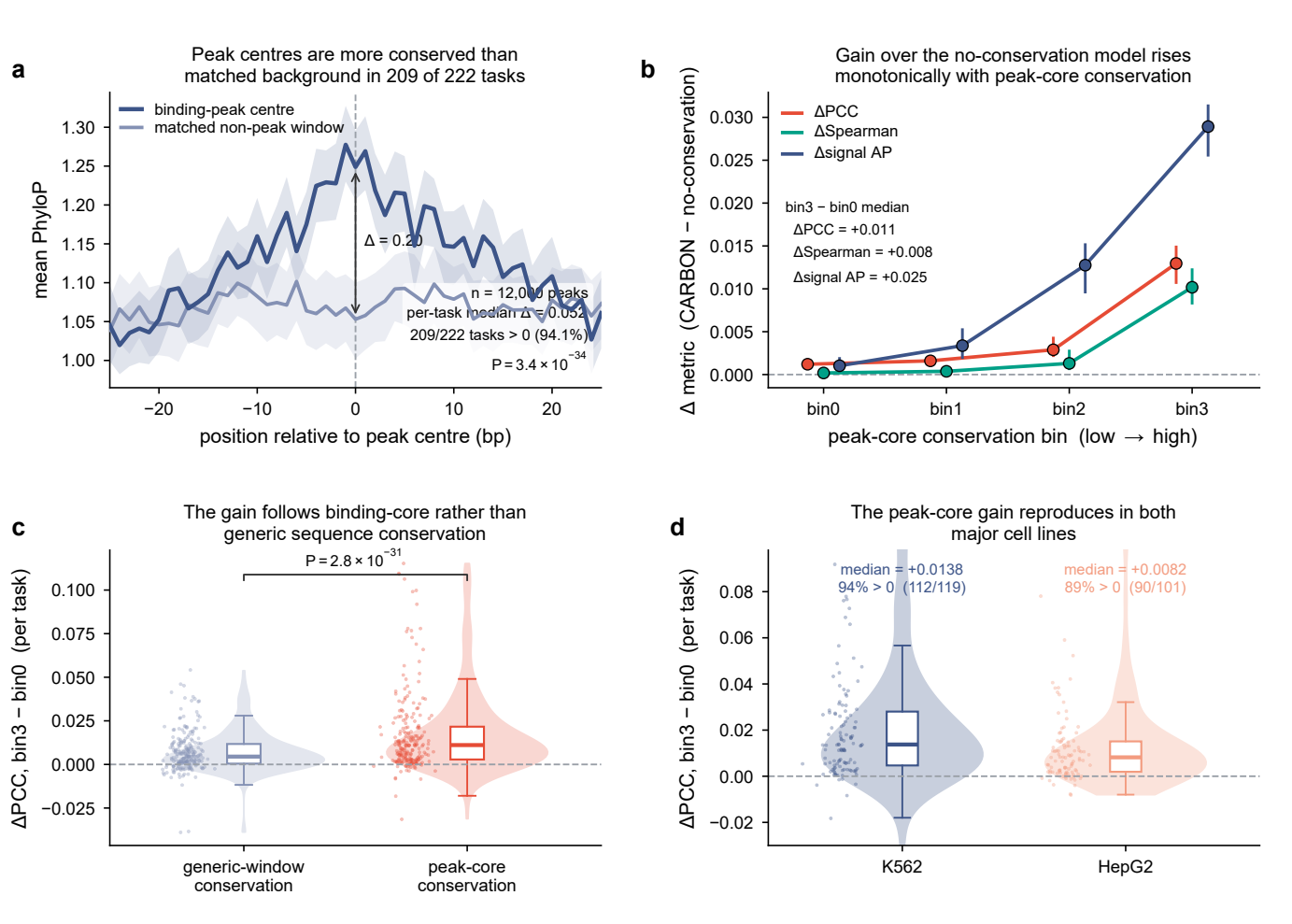

a
Peak centres are more conserved than matched background in 209 of 222 tasks
binding-peak centre
matched non-peak window
mean PhyloP
Δ = 0.20
n = 12,000 peaks
per-task median Δ = 0.052
209/222 tasks > 0 (94.1%)
P = 3.4 × 10−34
position relative to peak centre (bp)
b
Gain over the no-conservation model rises monotonically with peak-core conservation
ΔPCC
ΔSpearman
Δsignal AP
bin3 − bin0 median
ΔPCC = +0.011
ΔSpearman = +0.008
Δsignal AP = +0.025
Δ metric (CARBON − no-conservation)
bin0
bin1
bin2
bin3
peak-core conservation bin (low → high)
c
The gain follows binding-core rather than generic sequence conservation
P = 2.8 × 10−31
ΔPCC, bin3 − bin0 (per task)
generic-window conservation
peak-core conservation
d
The peak-core gain reproduces in both major cell lines
median = +0.0138
94% > 0 (112/119)
median = +0.0082
89% > 0 (90/101)
ΔPCC, bin3 − bin0 (per task)
K562
HepG2